



ANL Staking Protocol

WHITE PAPER

ANL staking protocol with a triple reward stream: ANL + XNT + CAPY
X1 Network

1. Summary

ANL Staking Protocol is a non-custodial staking protocol for ANL tokens running on **X1 Network**. A participant locks ANL for a self-declared period (minimum 7 days) and receives three reward streams: a **fixed APY paid in ANL** from a finite, segregated pool of 200,000,000 ANL; a **daily-accrued share of XNT rewards** generated by a dedicated X1 validator run by the protocol team; and a **CAPY bonus** — a share of a segregated pool of 20,000,000 CAPY, credited at the moment a position is claimed, proportionally to the ANL reward paid.

The protocol offers two programs: **Genesis** — with an elevated APY that depends on the time window in which the participant joined (up to 20% per year), and **Flexible** — with a fixed 8% per year regardless of entry time. In both programs the staking period is declared by the participant, and the rate assigned to a position at opening is **immutable for the entire life of that position**.

A position is settled **on the day the declared period ends**: the participant receives the ANL reward and all XNT accrued to the position at once; at that same moment the CAPY bonus is calculated and reserved, to be claimed with a separate instruction. In the Flexible program, early exit returns the principal in full but forfeits all rewards; **Genesis positions are locked until the end of the declared period** — with no early-exit option.

Protocol parameters

Parameter	Value
Network	X1 Network (x1.xyz)
Staked token	ANL
Reward tokens	ANL (fixed APY) + XNT (daily accrual) + CAPY (bonus at claim)
ANL reward pool	200,000,000 ANL (finite, segregated)
CAPY reward pool	20,000,000 CAPY (finite, segregated)
Programs	Genesis (APY windows) · Flexible (8%)
Staking period	declared by the participant, minimum 7 days
Genesis APY	20% / 15% / 8% — depending on the entry window
Flexible APY	8% — regardless of window
Daily XNT pool split	65% Genesis programs · 35% Flexible
XNT validator	“ANL Staking Protocol” — live; team self-stake (the minimum required by X1 Network)
Reward payout	Genesis: XNT every 30 days (02:00 UTC), ANL + principal at period end · Flexible: everything on the day the period ends · CAPY: separate claim after the position is claimed

Parameter	Value
Minimum deposit	1 ANL
Protocol year	365 days; periods counted in full days
Early exit	Flexible — yes (rewards forfeited) · Genesis — none, locked until period end
Production launch	pilot with a 5,000,000 ANL pool (from the 200M reward pool), after an AI-assisted audit

2. Vision and context

X1 Network is a high-performance blockchain compatible with the SVM tooling ecosystem. Its security and throughput are provided by validators, who are rewarded for their work in the native token XNT — paid out after every epoch, i.e. in practice daily. ANL Staking Protocol connects the community of ANL holders with the real work of a validator: instead of promising yield from unlimited emission, the protocol shares the validator's **real, daily revenue** and pays ANL and CAPY rewards from **predetermined, finite pools**. It is a model you can calculate to the end — with no hidden inflation and no dependency on new participants joining.

Core principle

User principal, the ANL reward pool, the XNT pool and the CAPY pool are held in four separate vaults. The program never pays principal out of a reward vault or rewards out of the principal vault — an invariant hard-coded and covered by automated tests.

3. Reward tokenomics

ANL rewards come from a segregated pool of **200,000,000 ANL**, deposited in the Reward Vault. The pool acts as a liquidity reservoir: each position accrues rewards at its fixed, assigned annual rate, and payouts reduce the reservoir. A finite pool means a natural horizon for the program — the protocol does not mint new ANL.

XNT rewards come from the **revenue of the X1 validator** run by the protocol team, operating under the name **“ANL Staking Protocol”**. The validator is **already live and working in the network** — it receives XNT after each epoch, so the protocol has XNT on hand and can accrue it to positions **daily**, without waiting for annual settlements.

The team maintains the validator's **own self-stake at the minimum required by X1 Network**. A larger total stake strengthens the validator's position in the network and increases the rewards it receives — including through the X1 Foundation delegation program, in which well-staked, stable validators receive additional delegation. Every additional XNT of revenue goes in full to the daily distribution pool for stakers. The validator's own deposit is the team's skin in the game — it is **never distributed to participants**; only generated revenue is distributed.

CAPY rewards come from a segregated pool of **20,000,000 CAPY**, deposited in the Capy Vault. CAPY is a **bonus linked to the ANL stream**: the larger the share of the ANL reward pool a position claims, the proportionally larger its share of the available CAPY (mechanics in

section 9). The pool is finite — the protocol does not mint new CAPY; the vault may be additionally funded by team decision, which raises the bonus rate for future claims.

Validator by the numbers — as of 31 August 2026

Status: active (100/100) · total stake: over 1,000,000 XNT · commission: 10% · skip rate: 0.00% · block production: 98%. Current data can be verified live at any time in X1 explorers under the name “ANL Staking Protocol”.



The “ANL Staking Protocol” validator on X1 — data verifiable in explorers (August 2026).

4. Three reward streams

Each position earns in parallel across two continuous streams — ANL and XNT — and, at the moment of claim, additionally receives a one-time CAPY bonus. The ANL and XNT streams are settled on the day the position's period ends; the CAPY bonus is calculated at claim and paid out with a separate instruction:

Trait	ANL stream	XNT stream	CAPY bonus
Source	finite 200M ANL pool	daily X1 validator revenue	finite 20M CAPY pool
Nature	fixed annual rate (APY)	proportional share of the daily pool	bonus proportional to the ANL reward
Accrual	continuous, by time and position rate	daily, proportional to stake	one-time, at position claim

Trait	ANL stream	XNT stream	CAPY bonus
Guarantee	yes — rate immutable per position	no — depends on validator work	after crediting — yes (reserved); before, depends on pool state
Payout	on the day the position's period ends	Genesis: every 30 days · Flexible: with ANL at period end	separate claim after the position is claimed

Immutable APY

The ANL rate assigned to a position at staking applies until the end of its life — regardless of the number of participants and later changes to program windows. Changes to protocol parameters never apply retroactively to open positions.

5. The Genesis program

Genesis rewards early entry into the protocol. The participant declares the staking period (minimum 7 days), and the APY depends on the time window in which they joined the program — counted from the protocol's public launch:

Entry window	Program days	APY (ANL)
Window 1 — early access	from day 0 to the end of day 30	20% per year
Window 2	from day 31 to the end of day 90	15% per year
Window 3 — standard	from day 91	8% per year

Genesis = commitment

Genesis positions are irrevocable: the declared period locks the principal to the very end — with no early-exit option. It is a deliberate trade: an elevated APY in exchange for a hard commitment that gives the protocol and community a predictable, on-chain-verifiable supply. In return, Genesis positions receive a payout of accrued XNT every 30 days — always at 02:00 UTC. The payout is executed by the protocol's automation, and the instruction is permissionless: after the deadline anyone can trigger it, so the payout does not depend on a single operator.

Window boundaries are unambiguous to the second: each window spans from its start to the start of the next, and switches always happen at a fixed hour, 02:00 UTC. Rate assignment is determined solely by the moment a position opens — a position opened in Window 1 keeps 20% per year for the entire declared period, even if it runs long after the window closes.

Example

A participant opens a Genesis position in Window 1 for 60 days with 100,000 ANL. Over 60 days the position accrues an ANL reward at 20% per year ($100,000 \times 20\% \times 60/365 \approx 3,288$ ANL) and a daily share of the XNT pool. On day 60 accrual ends, and the participant claims the ANL reward and all accrued XNT in a single instruction; at the moment of claim a CAPY bonus is additionally calculated (section 9), which the participant withdraws with a separate claim.

6. The Flexible program

Flexible is a simpler path for participants who join at any time and want a predictable, fixed rate: **8% per year regardless of entry window**. Position mechanics are identical to Genesis: the participant declares a period (minimum 7 days), the rate is immutable, and settlement happens on the day the period ends. Flexible positions take part in the daily XNT distribution within the 35% basket.

The difference between the programs thus comes down to the source of the rate: Genesis rewards early entry with an elevated APY, Flexible offers a fixed 8% always — including when the higher-rate Genesis windows are already closed.

7. Position lifecycle

The following rules apply equally in both programs:

- **Opening.** The participant chooses a program, amount (min. 1 ANL) and period (min. 7 days). The position receives the APY in effect at the moment of opening — permanently.
- **Accrual.** The ANL reward grows continuously at the position's rate; the XNT reward is credited daily, proportionally to stake (section 8).
- **End of period.** Once the declared period elapses, accrual of both streams stops. The position earns no more — even if the participant delays claiming.
- **Claim.** On the day the period ends, or at any later time, the participant receives at once: the ANL reward, accrued XNT and principal. At the moment of claim a CAPY bonus is calculated and reserved (section 9), paid out with a separate claim — with no deadline, it does not expire. Rewards accrued after the period ends do not expire — they simply stop growing.
- **Periodic XNT payout — Genesis only.** Accrued XNT for Genesis positions is paid out every 30 days, at epoch boundaries at 02:00 UTC; principal and the ANL reward remain locked until the end of the period. Flexible positions receive XNT only at final settlement.
- **Early exit — Flexible only.** A Flexible participant may withdraw principal before the end of the period at any time. The principal returns immediately and in full, but all of the position's rewards are forfeited: unaccrued and accrued ANL reward returns to the reward pool, and accrued XNT returns to the distribution pool. An exited position receives no CAPY bonus — the bonus is calculated from the ANL reward, which is forfeited. Genesis positions have no exit option — principal and rewards are claimed only after the period ends.
- **Multiple positions.** Each entry creates a separate position with its own rate, period and settlement.

Exit on day 59 of 60 — plainly

Example: a 60-day Flexible position exited on day 59 returns 100% of principal, but the ANL reward and all accrued XNT are forfeited in full, and no CAPY bonus is calculated. The protocol does not partially settle an unfinished period — a period is either honored or not. Before exiting a position, check in the app how many days remain and how much reward is already accrued. Genesis positions cannot be exited.

8. XNT rewards — the daily mechanism

The protocol's validator receives XNT after each X1 epoch. XNT accumulated over a given day goes into the XNT vault and is distributed the same day among active positions according to the following rules:

- **Basket split.** The daily XNT pool is split 65% for Genesis positions and 35% for Flexible positions.
- **Split within a basket.** Within a basket, each active position receives a share proportional to the size of its ANL stake relative to the sum of stakes of all active positions in that basket on that day.
- **An active position** is a position whose declared period is still running. Positions past their period end (awaiting claim) do not participate in the daily distribution.
- **Accounting.** Accrued XNT is credited to the position and accumulated. Genesis positions receive a payout of accrued XNT every 30 days; for Flexible positions the payout happens with the ANL reward — on the day the period ends.
- **Empty basket.** If on a given day a basket has no active position, its daily pool carries over to the next day within the same basket.

The size of the daily XNT pool depends on the validator's actual work (uptime, validator stake size, network parameters) and is **not guaranteed**. The protocol distributes 100% of the validator's generated revenue — no more and no less.

Daily accrual formula

For position i in basket k (Genesis: basket share $u_k = 0.65$; Flexible: $u_k = 0.35$) the daily XNT accrual is:

$$XNT_i = P_{day} \times u_k \times stake_i / \sum stake_k$$

where P_{day} is the XNT pool collected by the validator on that day, and $\sum stake_k$ is the sum of ANL across all active positions in the basket.



9. CAPY rewards — the bonus mechanism

The CAPY bonus is calculated **at the moment a position is claimed**, based on the ANL reward actually paid out then. It requires no additional action during the position's life — it is enough to honor the declared period and claim the rewards:

$$CAPY = ANL\ reward \times available\ CAPY / remaining\ ANL$$

where **available CAPY** is the Capy Vault balance minus amounts already reserved for other participants, and **remaining ANL** is 200,000,000 minus the sum of ANL rewards paid out so far.

- **Proportionality.** By claiming 1% of the remaining ANL reward pool, a position receives 1% of the currently available CAPY. The “CAPY per ANL” rate is, at any given moment, shared by all participants — regardless of program and position size.
- **Reservation = guarantee.** The calculated amount is immediately reserved in the protocol's accounting. From that moment no later payout to other participants can reduce it — the bonus waits to be claimed, risk-free and with no deadline.
- **Separate claim.** Reserved CAPY is paid out with a separate instruction at any time after the position is claimed — including immediately, in the same session. Reservations from multiple positions accumulate on the participant's profile and can be claimed together.

- **Pool limit.** The calculation never exceeds the available vault balance; when the CAPY pool is exhausted, subsequent claims receive only ANL and XNT.
- **Exit = no bonus.** An early exit of a Flexible position means forfeiting the ANL reward — and without an ANL reward, no CAPY bonus is calculated.

The size of the bonus is not guaranteed in advance — it depends on the Capy Vault balance and the sum of ANL rewards paid out at the moment of claim. Additional funding of the vault raises the bonus rate for future claims.

Example

Capy Vault: 20,000,000, reservations: 0. Of the ANL reward pool, 10,000,000 has been paid out so far, so remaining ANL = 190,000,000. A position claims 3,288 ANL of reward → bonus = $3,288 \times 20,000,000 / 190,000,000 \approx 346$ CAPY, immediately reserved and ready to be paid out with a separate claim.

10. Architecture and security

Four isolated vaults

Vault	Contents	Rule
Principal Vault	user principal	payouts only to position owners
Reward Vault (ANL)	pool of 200,000,000 ANL	payouts only as accrued rewards
XNT Vault	validator revenue	daily distribution per section 8
Capy Vault	pool of 20,000,000 CAPY	payouts only as reserved bonuses (section 9)

- **Principal untouchable.** Principal returns come solely from the Principal Vault; no one — not even the administration — can pay out a user's principal as a “reward”.
- **Mistake protection (ClaimFirst).** The contract will not allow closing a position with unclaimed due rewards — claim and payout are executed in a single transaction.
- **CAPY reservation guarantee.** Reserved bonuses are accounted globally; the Capy Vault will never pay out more than the sum of reservations, and calculated amounts cannot be taken by later claims.
- **Controlled launch.** The protocol launches paused, runs a pilot with a 5,000,000 ANL pool and 24/7 monitoring before fully funding the reward pool.
- **Audit.** The code undergoes an AI-assisted security audit, invariant tests and fuzzing (24h+) before production launch.

11. Roadmap

Period	Stage	Scope
07-08 / 2026	Implementation and tests	Rust + Anchor smart contract; 250+ test cases; invariant tests; third CAPY reward stream
09 / 2026	AI audit and X1 Testnet	AI-assisted security audit; full three-stream cycle on testnet; 24h+ fuzzing
10 / 2026	Mainnet Pilot	5,000,000 ANL pilot pool; controlled opening (launch paused); 24/7 monitoring
11-12 / 2026	Full launch	funding of the 200,000,000 ANL and 20,000,000 CAPY pools; staking dashboard; start of daily XNT accrual

12. Risk factors

- **Smart contract risk.** Despite tests, an audit and a pilot, bugs in the code cannot be ruled out. A staged launch limits the potential scale of damage.
- **Validator risk.** The size of XNT rewards depends on the uninterrupted, correct operation of the validator and the parameters of X1 Network; it is not guaranteed.
- **Variable CAPY rate.** The size of the CAPY bonus depends on the vault balance and the sum of ANL rewards paid out at the moment of claim; it is guaranteed only after being calculated and reserved. The CAPY pool is finite and may be exhausted.
- **Market risk.** The value of ANL, XNT and CAPY is subject to market fluctuations independent of the protocol. The fixed APY applies to token quantity, not value.
- **Network risk.** The protocol inherits the infrastructure risks of X1 Network (outages, changes to epoch parameters, upgrades).
- **ANL pool exhaustion.** The reward pool is finite; the protocol stops accepting new positions whose rewards would not be covered by the reservoir.

13. Glossary

Term	Meaning
Position	a single stake with its own amount, rate and period
Entry window	the range of days from protocol launch that determines the Genesis APY
Immutable APY	the rate assigned to a position at opening, unchanged until the end
XNT basket	a share of the daily XNT pool: 65% Genesis, 35% Flexible
Active position	a position whose declared period is still running
Claim	receiving ANL + XNT rewards and principal after the period ends; at this moment the CAPY bonus is calculated

Term	Meaning
CAPY bonus	a one-time CAPY calculation at position claim, proportional to the ANL reward paid
CAPY reservation	the accounting of the calculated bonus; from this moment the amount is guaranteed and awaits payout
CAPY claim (claim_capy)	a separate instruction to pay out reserved CAPY — at any time after the position is claimed
Exit	withdrawing principal before the period ends (Flexible only); rewards are forfeited, no CAPY bonus is calculated
Epoch	the X1 settlement cycle after which the validator receives XNT
Periodic XNT payout	the automatic payout of accrued XNT for Genesis positions every 30 days (02:00 UTC)

Changelog

Version	Date	Changes
1.0	07 / 2026	first publication
1.1	07 / 2026	Genesis program: funds locked for the entire declared period (no exit option) and periodic payout of accrued XNT every 30 days; production-launch pilot pool increased to 5,000,000 ANL (segregated from the 200M reward pool)
1.2	08 / 2026	third reward stream: CAPY bonus from a segregated 20,000,000 CAPY pool — calculated at position claim proportionally to the ANL reward, guaranteed by reservation, paid out with a separate claim; fourth Capy Vault; validator self-stake description unified to the minimum required by X1 Network

This document is informational and does not constitute investment advice or an offer within the meaning of applicable law. APY values refer to rewards denominated in ANL; XNT rewards depend on the validator's actual work and are not guaranteed in amount; the size of the CAPY bonus depends on the pool state at the moment of claim and is not guaranteed in advance. Participation in blockchain protocols involves risk, including the risk of loss of funds.